

Experiment 6: CloudSim Simulation with Shortest Job First Scheduling

Aim

To simulate a cloud environment using CloudSim and replace the default broker submission order with a custom Shortest Job First (SJF) scheduling strategy.

Tools Required

- Ubuntu VM
- OpenJDK
- Git
- Maven
- IntelliJ IDEA Community
- CloudSim source code

Detailed Procedure

Install the required Java, Git, and Maven tools.

```
sudo apt update
sudo apt install default-jdk git maven -y
```

Install IntelliJ IDEA Community. If snap is available, use:

```
sudo snap install intellij-idea-community --classic
```

Clone CloudSim and enter the project directory.

```
git clone https://github.com/Cloudslab/cloudsim.git
cd cloudsim
```

Open /home/<username>/cloudsim in IntelliJ IDEA as a Maven project.

Set the Project JDK to the installed OpenJDK. In the completed workflow, OpenJDK 25.0.4 was selected.

Build the project from the IntelliJ terminal.

```
mvn clean package -DskipTests
```

Verify that the Maven build finishes with BUILD SUCCESS.

Create SJFDatacenterBroker.java under org.cloudbus.cloudsim.examples. Extend DatacenterBroker and override submitCloudlets() so the pending Cloudlets are sorted by length before the parent broker submits them.

```
package org.cloudbus.cloudsim.examples;

import org.cloudbus.cloudsim.Cloudlet;
import org.cloudbus.cloudsim.DatacenterBroker;
import java.util.Comparator;

public class SJFDatacenterBroker extends DatacenterBroker {
    public SJFDatacenterBroker(String name) throws Exception { super(name); }
```

```
@Override
protected void submitCloudlets() {
    getCloudletList().sort(Comparator.comparingLong(Cloudlet::getCloudletLength));
    super.submitCloudlets();
}
}
```

Create SJFSimulation.java. Initialize CloudSim, create a datacenter and the custom broker, create one VM, and create Cloudlets with different lengths.

Use the current CloudSim API names. Submit the VM list with submitGuestList(), check success using getStatus() and Cloudlet.CloudletStatus.SUCCESS, use getGuestId(), and use getExecFinishTime() for the finish time.

```
broker.submitGuestList(vmList);

if (cloudlet.getStatus() == Cloudlet.CloudletStatus.SUCCESS) {
    System.out.println("Cloudlet " + cloudlet.getCloudletId()
        + " | Length: " + cloudlet.getCloudletLength()
        + " MI | VM: " + cloudlet.getGuestId()
        + " | Start: " + dft.format(cloudlet.getExecStartTime())
        + " | Finish: " + dft.format(cloudlet.getExecFinishTime()));
}
```

Create Cloudlets with lengths 40000 MI, 10000 MI, and 20000 MI and submit them initially in a non-SJF order so that the custom broker's sorting can be observed.

Run SJFSimulation.main() from IntelliJ.

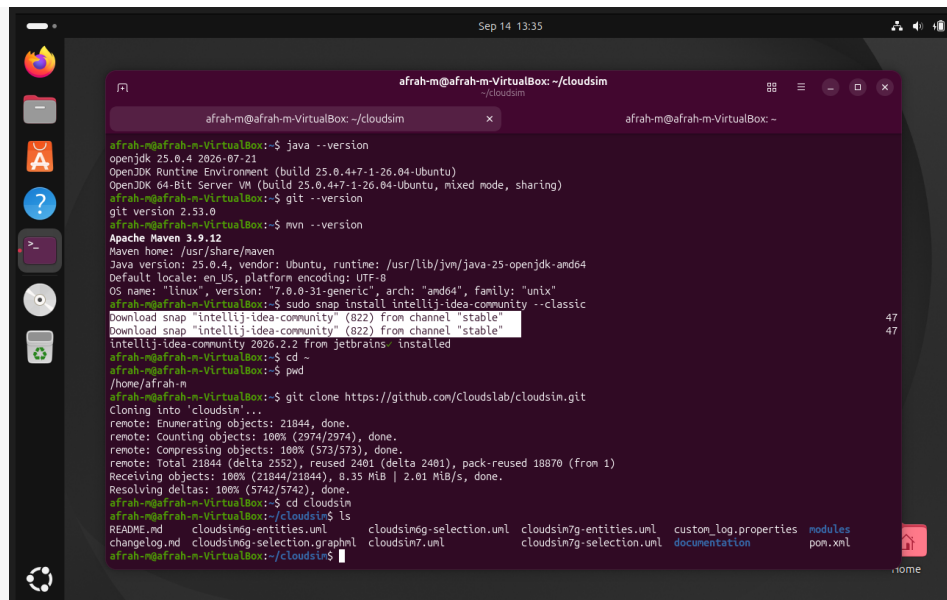
Verify that the completed output lists the shortest Cloudlet first, followed by the medium and longest Cloudlet. The successful run produced the order 10000 MI → 20000 MI → 40000 MI and exited with code 0.

Important Compatibility Note

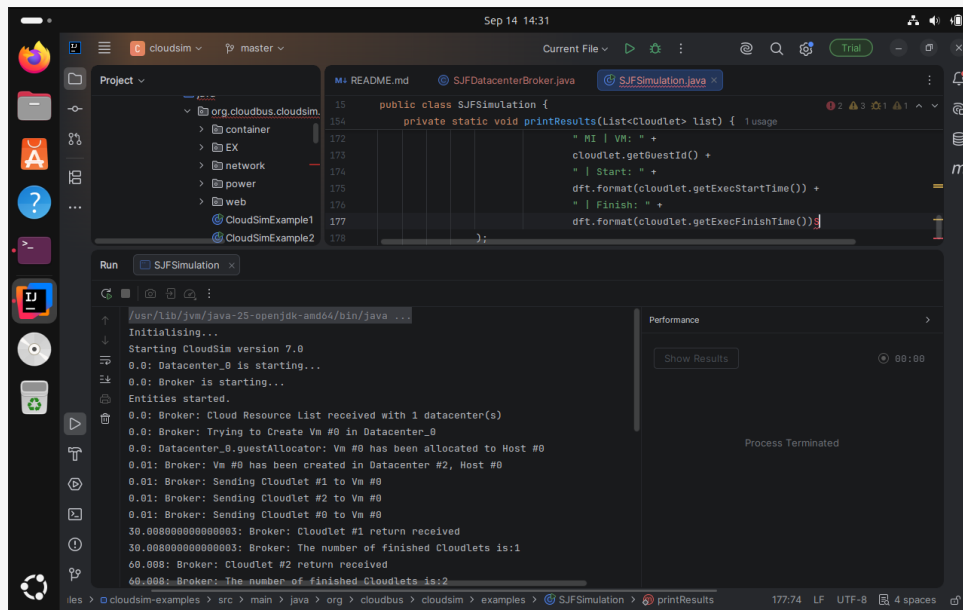
The original simulation code used older CloudSim APIs such as submitVmList(), getCloudletStatus(), Cloudlet.SUCCESS, getVmId(), and getFinishTime(). CloudSim 7.0.1 uses the current GuestEntity-oriented API, so the procedure above uses submitGuestList(), getStatus(), Cloudlet.CloudletStatus.SUCCESS, getGuestId(), and getExecFinishTime().

Output

```
Sep 14 13:26
afrah-m@afrah-m-VirtualBox: ~
afrah-m@afrah-m-VirtualBox: ~
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jdeps to provide /usr/bin/jdeps (jdeps) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jfr to provide /usr/bin/jfr (jfr) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jimage to provide /usr/bin/jimage (jimage) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jinfo to provide /usr/bin/jinfo (jinfo) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jlink to provide /usr/bin/jlink (jlink) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jmap to provide /usr/bin/jmap (jmap) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jmod to provide /usr/bin/jmod (jmod) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jnative2java to provide /usr/bin/jnative2java (jnative2java) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jps to provide /usr/bin/jps (jps) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jrunscript to provide /usr/bin/jrunscript (jrunscript) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jshell to provide /usr/bin/jshell (jshell) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jstack to provide /usr/bin/jstack (jstack) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jstat to provide /usr/bin/jstat (jstat) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jstatd to provide /usr/bin/jstatd (jstatd) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jwebserver to provide /usr/bin/jwebserver (jwebserver) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/serialver to provide /usr/bin/serialver (serialver) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jhsdb to provide /usr/bin/jhsdb (jhsdb) in auto mode
Setting up maven (3.9.12-1)...
update-alternatives: using /usr/share/maven/bin/mvn to provide /usr/bin/mvn (mvn) in auto mode
Setting up default-jre (2:1.25-77)...
Setting up default-jdk-headless (2:1.25-77)...
Setting up openjdk-25-jdk-and64 (25.0.4+7-1-26.04-Ubuntu)...
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jconsole to provide /usr/bin/jconsole (jconsole) in auto mode
Setting up default-jdk (2:1.25-77)...
afrah-m@afrah-m-VirtualBox: ~$ java --version
openjdk 25.0.4 2026-07-21
OpenJDK Runtime Environment (build 25.0.4+7-1-26.04-Ubuntu)
OpenJDK 64-Bit Server VM (build 25.0.4+7-1-26.04-Ubuntu, mixed mode, sharing)
afrah-m@afrah-m-VirtualBox: ~$ git --version
git version 2.53.0
afrah-m@afrah-m-VirtualBox: ~$ mvn --version
Apache Maven 3.9.12
Maven home: /usr/share/maven
Java version: 25.0.4, vendor: Ubuntu, runtime: /usr/lib/jvm/java-25-openjdk-and64
Default locale: en_US, platform encoding: UTF-8
OS name: "linux", version: "7.0.0-31-generic", arch: "amd64", family: "unix"
afrah-m@afrah-m-VirtualBox: ~$
afrah-m@afrah-m-VirtualBox: ~
afrah-m@afrah-m-VirtualBox: ~
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jrunscript to provide /usr/bin/jrunscript (jrunscript) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jshell to provide /usr/bin/jshell (jshell) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jstack to provide /usr/bin/jstack (jstack) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jstat to provide /usr/bin/jstat (jstat) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jstatd to provide /usr/bin/jstatd (jstatd) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jwebserver to provide /usr/bin/jwebserver (jwebserver) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/serialver to provide /usr/bin/serialver (serialver) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jhsdb to provide /usr/bin/jhsdb (jhsdb) in auto mode
Setting up maven (3.9.12-1)...
update-alternatives: using /usr/share/maven/bin/mvn to provide /usr/bin/mvn (mvn) in auto mode
Setting up default-jre (2:1.25-77)...
Setting up default-jdk-headless (2:1.25-77)...
Setting up openjdk-25-jdk-and64 (25.0.4+7-1-26.04-Ubuntu)...
update-alternatives: using /usr/lib/jvm/java-25-openjdk-and64/bin/jconsole to provide /usr/bin/jconsole (jconsole) in auto mode
Setting up default-jdk (2:1.25-77)...
afrah-m@afrah-m-VirtualBox: ~$ java --version
openjdk 25.0.4 2026-07-21
OpenJDK Runtime Environment (build 25.0.4+7-1-26.04-Ubuntu)
OpenJDK 64-Bit Server VM (build 25.0.4+7-1-26.04-Ubuntu, mixed mode, sharing)
afrah-m@afrah-m-VirtualBox: ~$ git --version
git version 2.53.0
afrah-m@afrah-m-VirtualBox: ~$ mvn --version
Apache Maven 3.9.12
Maven home: /usr/share/maven
Java version: 25.0.4, vendor: Ubuntu, runtime: /usr/lib/jvm/java-25-openjdk-and64
Default locale: en_US, platform encoding: UTF-8
OS name: "linux", version: "7.0.0-31-generic", arch: "amd64", family: "unix"
afrah-m@afrah-m-VirtualBox: ~$ sudo snap install intelliij-idea-community --classic
Download snap "intelliij-idea-community" (822) from channel "stable"
Download snap "intelliij-idea-community" (822) from channel "stable"
intelliij-idea-community 2026.2.2 from jetbrains: installed
afrah-m@afrah-m-VirtualBox: ~$
```

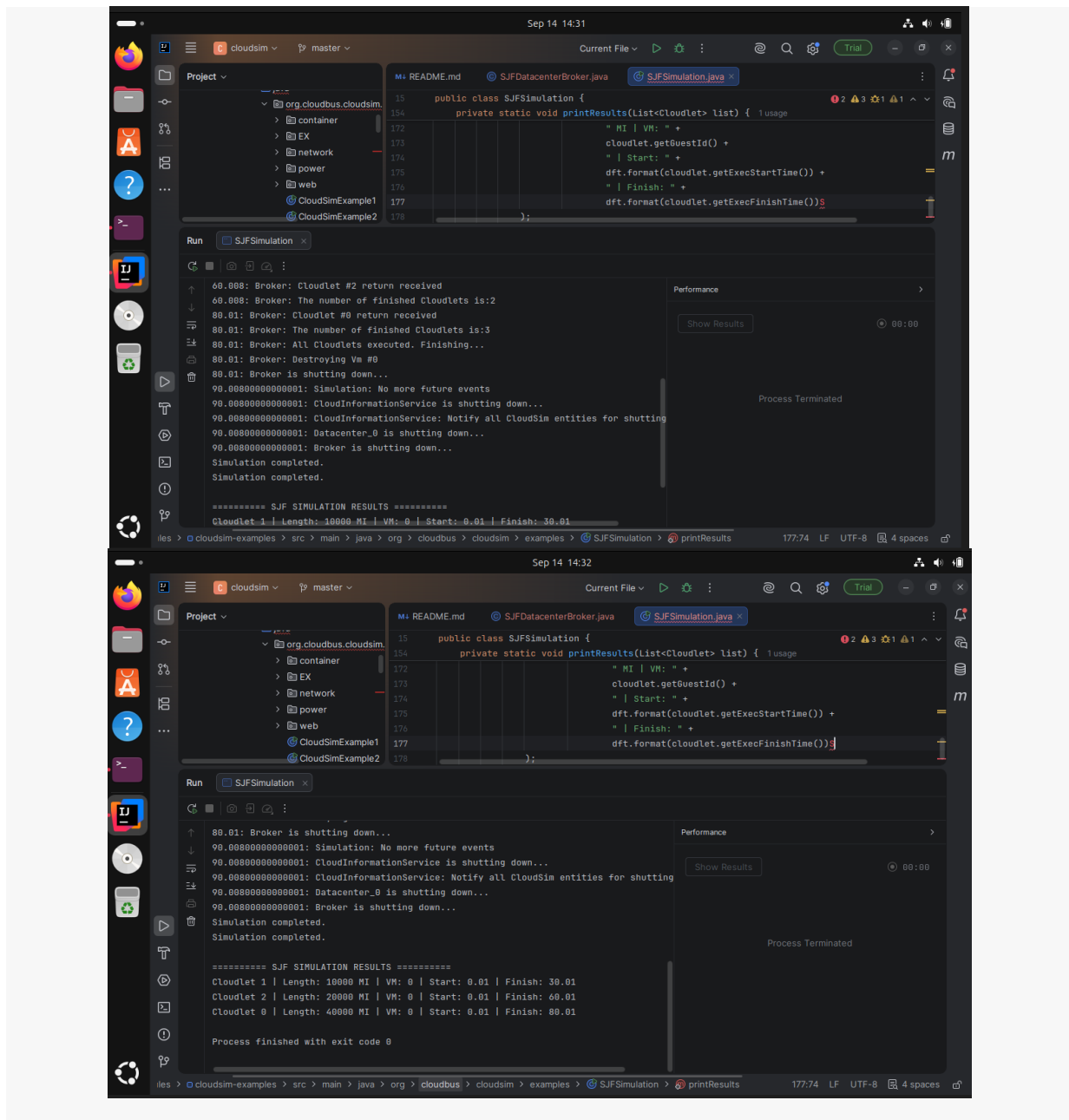


```
afrah-m@afrah-m-VirtualBox: ~/cloudsim
afrah-m@afrah-m-VirtualBox:~/cloudsim$ java --version
openjdk 25.0.4 2026-07-21
OpenJDK Runtime Environment (build 25.0.4-71-26.04-Ubuntu)
OpenJDK 64-Bit Server VM (build 25.0.4-71-26.04-Ubuntu, mixed mode, sharing)
afrah-m@afrah-m-VirtualBox:~/cloudsim$ git --version
git version 2.53.0
afrah-m@afrah-m-VirtualBox:~/cloudsim$ mvn --version
Apache Maven 3.9.12
Maven home: /usr/share/maven
Java version: 25.0.4, vendor: Ubuntu, runtime: /usr/lib/jvm/java-25-openjdk-amd64
Default locale: en_US, platform encoding: UTF-8
OS name: "linux", version: "7.0.0-31-generic", arch: "amd64", family: "unix"
afrah-m@afrah-m-VirtualBox:~/cloudsim$ sudo snap install intelliij-idea-community --classic
Download snap "intelliij-idea-community" (822) from channel "stable"
Download snap "intelliij-idea-community" (822) from channel "stable"
IntelliJ-idea-community 2026.2.2 from JetBrains: installed
afrah-m@afrah-m-VirtualBox:~/cloudsim$ cd -
/home/afrah-m
afrah-m@afrah-m-VirtualBox:~/cloudsim$ git clone https://github.com/CloudSim/cloudsim.git
Cloning into 'cloudsim'...
remote: Enumerating objects: 21844, done.
remote: Counting objects: 100% (2074/2074), done.
remote: Compressing objects: 100% (573/573), done.
remote: Total 21844 (delta 2552), reused 2401 (delta 2401), pack-reused 18870 (from 1)
Receiving objects: 100% (21844/21844), 8.35 MiB | 2.01 MiB/s, done.
Resolving deltas: 100% (5742/5742), done.
afrah-m@afrah-m-VirtualBox:~/cloudsim$ cd cloudsim
afrah-m@afrah-m-VirtualBox:~/cloudsim$ ls
README.md  cloudsim-entities.uml  cloudsim-selection.uml  cloudsim7-entities.uml  custom_log.properties  modules
changeLog.md  cloudsim7-selection.graphml  cloudsim7.uml  cloudsim7-selection.uml  documentation  pom.xml
afrah-m@afrah-m-VirtualBox:~/cloudsim$
```



```
public class SJFSimulation {
    private static void printResults(List<Cloudlet> list) {
        // MI | VM: " +
        cloudlet.getGuestId() +
        " | Start: " +
        dft.format(cloudlet.getExecStartTime()) +
        " | Finish: " +
        dft.format(cloudlet.getExecFinishTime())
    }
}

Run SJFSimulation
/usr/lib/jvm/java-25-openjdk-amd64/bin/java ...
Initialising...
Starting CloudSim version 7.0
0.0: Datacenter_0 is starting...
0.0: Broker is starting...
Entities started.
0.0: Broker: Cloud Resource List received with 1 datacenter(s)
0.0: Broker: Trying to Create Vm #0 in Datacenter_0
0.0: Datacenter_0.guestAllocator: Vm #0 has been allocated to Host #0
0.01: Broker: Vm #0 has been created in Datacenter #2, Host #0
0.01: Broker: Sending Cloudlet #1 to Vm #0
0.01: Broker: Sending Cloudlet #2 to Vm #0
0.01: Broker: Sending Cloudlet #0 to Vm #0
30.000000000000003: Broker: Cloudlet #1 return received
30.000000000000003: Broker: The number of finished Cloudlets is:1
60.000: Broker: Cloudlet #2 return received
60.000: Broker: The number of finished Cloudlets is:2
Performance
Show Results
Process Terminated
```



Result

CloudSim was built successfully and the custom SJF broker scheduled the Cloudlets from shortest to longest, demonstrating Shortest Job First scheduling.